order17 RESULTS
PREREG sha256 d3938e1c034b961d3577d3253bb3d836f189db7542fea0923fe2d4a2e313e049 (written and hashed before numerics).

Verdicts by the preregistered rules (recorded first)
test	value	verdict
C0 control (fixed-M law on A0)	r0 = 1.3e−15	PASS
K1 gate (Cor. R1 + line terms + P)	e = 1.0e−15	PASS
T1 primary, weight ½, full A1c	r = 0.9991	**FAIL**
T2 w=0, w=1, full	r = 0.9994, 0.9992	FAIL
T1/T2 on Π+A1c (w=½,0,1)	r ≈ 0.996	FAIL
T1/T2 on Π−A1c	r ≈ 0.99999	FAIL (vacuous by PREREG)

Predictions: K1 PASS (correct), T1 FAIL (recorded), Π+ FAIL (recorded).

Post-hoc reading (disclosed; not a change of verdict)
‖A1c‖_F = 4.0e−14 against ‖A1_lit‖_F = 63.35 (relative 6e−16): A1c is zero to rounding at all 216 points and at their −1/x images, in both sectors. The relative residual r is a ratio of rounding noise, so T1/T2 are non-discriminating (same zero-target defect as order13 W2d). Diagnostics (scripts/diag_posthoc.py): ‖Π+(A1_lit − A1_lines)‖ = 2.4e−14, ‖Π+ A1_pred‖ = 8.7e−15, ‖Π− A0‖ = 1.4e−15.

Category A (new, proofs in NOTES / paper v5)
- Prop. Z: the Zwegers law of $\hat{H}$, $\hat{G}_O$ is exactly $(35^{-1/2}M, T)$, $\chi = 1$ — Category A (Zwegers Cor. 2.9 + verbatim match with the symbol rules + exact closure). App. D item sh2 is no longer needed as evidence.
- Thm. R: the completed P-odd series has $A_j = 0$ for all j at every rational x.
- Thm. E (found from the zero target, proved afterwards): A1(S^E) = 0 for every companion and rational x (the ++ and −− quadrant boundary terms cancel class by class because $\tilde{B}1$ is odd).
- Hence A1_lit = A1(lines) + A1(S^o) exactly, and **A1 minus the explicit completion and line terms is identically 0**. A weight-½ (or any) fixed-M law for it holds only vacuously.

Paper
RTA_FOUNDATION_BOUND_v5.pdf built (see NOTES).

order17 NOTES (disclosures)
- Order: PREREG written and hashed (d3938e1c…e049) → k1t1.py hashed in scripts/SHA256SUMS → run → verdicts recorded in RESULTS → post-hoc diag_posthoc.py (hashed after).
- T1 primary FAIL is recorded by the preregistered relative-residual rule. The target A1c is identically zero (‖A1c‖ = 4e−14), so r ≈ 1 is noise/noise; the rule was not designed for a zero target (same defect as order13 W2d). Verdict unchanged; reading is post-hoc.
- PREREG predicted Π+A1c = ½A1(S^E) ≠ 0 and T2 FAIL on Π+. In fact ½A1(S^E) = 0. Theorem E (A1(S^E) = 0) was found from this zero and proved afterwards (quadrant-pair cancellation with odd $\tilde{B}1$). It is Category A but post-hoc in origin.
- Part 2: Zwegers law ↔ $(35^{-1/2}M, T)$, χ=1 is Category A by citation (Zwegers thesis Cor. 2.9) plus the verbatim match with the symbol S/T rules and the exact closure in S_rows/T_rows. Residual dependence: the exact printed form of Cor. 2.9; numerics of orders 5/6/11 corroborate. No further certificate needed.
- Theorem R relies on the Schwartz property of the completed odd kernel, argued via Zwegers' convergence estimate (noted as a problem in the paper NOTES_v5).
- The K1 gate shares the phase code (order2 lib2.phase_ints) with order3 parts; the formulas (wall sawtooth vs piece decomposition) are independent.
- No new Residual–Resonant map; E1–E3 OPEN. Framing (local mock / non-local quantum modular; "residual geometric bridge") remains opinion; no bridge claim.
- Runs: nice, OMP_NUM_THREADS=1, peak well under 3 GB, ~3 min. No push, no sends. New files only.